LESSON 5.6

Represent Three-Dimensional Figures in Two Dimensions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a solid by using its net.

WHY THIS MATTERS Your class is designing custom wrapping paper for time capsule boxes. To figure out how much paper you need, you have to find the total surface area of each box. The trick? Unfold the box into a net so you can see every face laid flat! One time capsule box is 6 inches long, 4 inches wide, and 2 inches tall. How much wrapping paper covers the whole box?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Surface area *Spanish: Área de superficie*	The total area of all the flat sides of a solid.	A box 6 × 4 × 2: SA = 2(24) + 2(12) + 2(8) = 88 in² — like wrapping paper needed
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	unfolded box
Face *Spanish: Cara*	One flat side of a solid shape.	a cube has 6 faces
Two-dimensional *Spanish: Bidimensional*	A flat shape with length and width, but no thickness.	a net on paper
Edge *Spanish: Arista*	The line where two flat sides of a solid meet.	A cube has 12 edges — the lines where two flat faces connect
Pyramid *Spanish: Pirámide*	A solid with one base and triangular sides that meet at a point.	square base + 4 triangles

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Net for a box 5 cm × 3 cm × 2 cm.

1. Bottom and top: 5 cm by 3 cm.
2. Front and back: 5 cm by 2 cm.
3. Two sides: 3 cm by 2 cm.

Answer: 6 rectangles, 3 matching pairs

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller box: 4 long, 3 wide, 2 tall. How many pairs of faces does it have?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This solid has _____ faces shaped like _____ .
- The net needs _____ flat pieces because _____ .
- I attached each _____ to a side of the _____ .
- When the net folds up, the _____ meets the _____ .

Word bank net · face · fold · unfold · base · pyramid · prism · cube · triangle · square · rectangle · two-dimensional · three-dimensional

WATCH OUT!

A common mistake

A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in × 4 in × 5 in, a student might add just one face from each pair: (9×4) + (9×5) + (4×5) = 36 + 45 + 20 = 101 in², but the net actually has two of each face, so the correct total is 2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 in²

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MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each concept about rectangular prisms.

_____ Top & Bottom faces	(A) l × w (two of them)
_____ Front & Back faces	(B) l × h (two of them)
_____ Left & Right faces	(C) w × h (two of them)
_____ Surface area formula	(D) SA = 2lw + 2lh + 2wh
_____ Volume formula	(E) V = l × w × h
_____ Number of faces	(F) 6 faces total

STEPS TO TRY

1. Each face pair uses two of the three dimensions: top/bottom skips height, front/back skips width, left/right skips length.
2. Match by asking which two edges touch that face — the top and bottom faces touch the length and width.
3. Start with 'Top & Bottom faces': that's the l × w pair. Then remember SA adds face areas while V multiplies l × w × h.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A time capsule box is a rectangular prism with l = 6 in, w = 4 in, h = 2 in. What is its surface area?

(A) 44 in²
(B) 48 in²
(C) 88 in²
(D) 88 in³

STEPS TO TRY

1. Surface area covers all 6 faces of the 6 × 4 × 2 box — it's not the same as the space inside.
2. Use SA = 2lw + 2lh + 2wh: find the three different face areas, double each, and add.
3. Start with the top face: 6 × 4 = 24 in². Then find 6 × 2 and 4 × 2 before doubling and adding.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

How many faces does a rectangular prism have?

(A) 4
(B) 6
(C) 8
(D) 12

STEPS TO TRY

1. Picture a cereal box in your hands. Turn it: there's a front, a back, two sides, a top, and a bottom.
2. Count faces in opposite pairs — every face on a rectangular prism has a matching partner across from it.
3. Count the pairs: top/bottom is one pair, front/back is another, left/right is the third. Multiply pairs by 2.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Surface area is measured in which type of unit?

(A) Cubic units (in³, cm³, ft³)

(B) Square units (in², cm², ft²)

(C) Linear units (in, cm, ft)

(D) No units needed

STEPS TO TRY

1. Surface area measures the flat covering on the outside of a solid — like wrapping paper.
2. Flat 2D measurements use square units. Save cubic units for filling space and plain units for length.
3. Imagine covering one face with 1-inch squares. Each square is 1 in² — that's the kind of unit surface area counts.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

SENTENCE STARTER The ___ faces contribute the most area (___ in²) because their dimensions ___ × ___ = ___ are the largest. The total SA is ___ + ___ + ___ = ___ in².

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help